Parallel & Perpendicular Lines

DSAT Math Practice • 25 Questions • Mixed Difficulty

Total Questions: 25 | **Multiple Choice:** 21 | **Grid-In (SPR):** 4

Question 1 **[Easy]**

Line *j* has a slope of 4. What is the slope of any line parallel to line *j*?

- **A)** 4
- **B)** -4
- **C)** 1/4
- **D)** -1/4

Question 2 **[Easy]**

Line *r* has a slope of -2/5. What is the slope of a line perpendicular to line *r*?

- **A)** 2/5
- **B)** -2/5
- **C)** 5/2
- **D)** -5/2

Question 3 **[Easy]**

Which of the following lines is parallel to y = 3x - 7?

- **A)** y = -3x + 7
- **B)** y = 3x + 12
- **C)** y = (1/3)x - 7
- **D)** y = -3x - 7

Question 4 **[Medium]**

A line in the xy-plane has the equation 6x + 2y = 10. Which of the following lines is perpendicular to this line?

- **A)** y = -3x + 5
- **B)** y = 3x - 1
- **C)** y = (1/3)x + 2
- **D)** y = (-1/3)x + 4

Question 5 **[Medium]**

Line *t* passes through (2, 5) and is parallel to the line y = -4x + 1. What is the equation of line *t*?

- **A)** y = -4x + 13
- **B)** y = -4x - 3
- **C)** y = (1/4)x + 13
- **D)** y = 4x + 13

Question 6 **[Medium]**

In the xy-plane, line *n* passes through (6, 1) and is perpendicular to the line y = (2/3)x + 5. What is the y-intercept of line *n*?

- **A)** (0, 10)
- **B)** (0, -8)
- **C)** (0, 7)
- **D)** (0, -10)

Question 7 **[Medium]**

The equation 8x - 4y = 20 defines a line in the xy-plane. Which of the following equations defines a line that is parallel to this line?

- **A)** y = -2x + 3
- **B)** y = 2x + 7
- **C)** y = (1/2)x - 5
- **D)** y = -1/2 x + 1

Question 8 **[Medium]**

Line *v* is defined by 3x + 9y = 27. Line *w* is perpendicular to line *v* and passes through (0, -4). What is the equation of line *w*?

- **A)** y = 3x - 4
- **B)** y = -3x - 4
- **C)** y = (1/3)x - 4
- **D)** y = -(1/3)x - 4

Question 9 **[Medium]**

Student-Produced Response

Two lines in the xy-plane are defined by y = kx + 3 and 2x + 8y = 16. If the two lines are parallel, what is the value of k?

Question 10 **[Medium]**

In the xy-plane, which of the following is the equation of a line perpendicular to $y = 5$ that passes through $(3, -2)$?

- **A)** $x = 3$
- **B)** $y = -2$
- **C)** $x = -2$
- **D)** $y = 5x - 2$

Question 11 **[Medium]**

A line passes through $(4, -1)$ and $(4, 7)$. Which of the following lines is perpendicular to this line?

- **A)** $x = 2$
- **B)** $y = 4x - 1$
- **C)** $y = 3$
- **D)** $x = -1$

Question 12 **[Medium]**

Line *a*: $5x - 10y = 30$. Line *b*: $y = mx + 2$. If lines *a* and *b* are perpendicular, what is the value of m?

- **A)** -2
- **B)** -1/2
- **C)** 1/2
- **D)** 2

Question 13 **[Medium]**

The line with equation $y = (3/4)x + 6$ is perpendicular to line *g* in the xy-plane. Line *g* passes through the point $(3, 1)$. What is the equation of line *g*?

- **A)** $y = -(4/3)x + 5$
- **B)** $y = (4/3)x + 5$
- **C)** $y = -(4/3)x - 3$
- **D)** $y = (3/4)x + 5$

Question 14 **[Easy]**

Two distinct lines in the xy-plane have slopes m■ and m■ respectively. If m■ · m■ = -1, which of the following must be true?

- **A)** The lines are parallel.
- **B)** The lines are perpendicular.
- **C)** The lines intersect at the origin.
- **D)** The lines have the same y-intercept.

Question 15 **[Hard]**

Student-Produced Response

Line *f* is parallel to the line $2x - 7y = 14$ and passes through (-7, 0). What is the y-intercept of line *f*?

Question 16 **[Hard]**

In the xy-plane, the line $3x + 4y = 24$ is perpendicular to line *h*. If line *h* passes through the point (3, 6), what is the x-intercept of line *h*?

- **A)** (-3/2, 0)
- **B)** (3/2, 0)
- **C)** (-6, 0)
- **D)** (6, 0)

Question 17 **[Hard]**

A triangle in the xy-plane has vertices at P(0, 0), Q(8, 0), and R(4, 6). Which of the following equations represents a line that contains the altitude from vertex R to side PQ?

- **A)** $x = 4$
- **B)** $y = 6$
- **C)** $y = (2/3)x$
- **D)** $x = 6$

Question 18 **[Hard]**

In the xy-plane, line *s* passes through (1, 3) and (5, 11). Line *u* is perpendicular to line *s* and passes through the midpoint of the segment from (1, 3) to (5, 11). What is the y-intercept of line *u*?

- **A)** (0, 17/2)
- **B)** (0, 8)
- **C)** (0, 15/2)
- **D)** (0, 29/4)

Question 19 **[Hard]**

Student-Produced Response

Line *c* has slope $(a + 2)/3$, and line *d* has slope $3/(a - 4)$, where a is a positive constant. If lines *c* and *d* are perpendicular, what is the value of a?

Question 20 **[Hard]**

In the xy-plane, a line parallel to $4x + 3y = 12$ passes through (3, 8) and (p, 4). What is the value of p?

- **A)** 0
- **B)** 3
- **C)** 6
- **D)** 9

Question 21 **[Medium]**

The equation of line *e* is kx - 3y = 9. If line *e* is parallel to the line y = (5/3)x + 2, what is the value of k?

- **A)** 3
- **B)** 5
- **C)** -3
- **D)** -5

Question 22 **[Medium]**

Line *q* passes through (-1, 4) and is perpendicular to the line passing through (0, 0) and (3, -6). What is the equation of line *q*?

- **A)** y = (1/2)x + 9/2
- **B)** y = (1/2)x + 4
- **C)** y = -2x + 2
- **D)** y = 2x + 6

Question 23 **[Hard]**

Two perpendicular lines intersect at (2, 5). If one line has the equation y = 3x - 1, which of the following could be the equation of the other line?

- **A)** y = -(1/3)x + 17/3
- **B)** y = 3x + 5
- **C)** y = (1/3)x + 13/3
- **D)** y = -(1/3)x - 17/3

Question 24 **[Hard]**

Student-Produced Response

In the xy-plane, the line ax + by = 1 is perpendicular to the line 4x - 6y = 3 and passes through the point (2, 1). What is the value of a + b?

Question 25 **[Medium]**

A rectangle in the xy-plane has one side along the line y = (1/2)x + 3. Which of the following could be the equation of a line containing an adjacent side of the rectangle?

- **A)** y = (1/2)x - 4
- **B)** y = -2x + 9
- **C)** y = 2x + 1
- **D)** y = -(1/2)x + 3
